Shortcut learning from image artifacts, region-of-interest masking and its alternatives in medical image classification: a review of the 2021–2026 literature

Literature review for the MSc thesis *Where the Shortcut Sits*

October 2026

Abstract

This review collects the peer-reviewed work published from January 2021 to October 2026 that bears on one question: when an image artifact that is correlated with the diagnosis lies inside or outside the region of interest (ROI), does restricting the classifier to the ROI remove the shortcut, and what should be used instead? We included 70 studies from journals and conferences of recognised standing (Nature, Lancet and Science families, Medical Image Analysis, MICCAI, CVPR, ECCV, NeurIPS, ICML, ICLR and others), verified each against its publisher or proceedings page and excluded preprints and workshop papers. The literature establishes firmly that medical classifiers exploit acquisition signals, devices, markings and demographic correlates; it documents artifacts per modality, mostly in dermoscopy and chest radiography; it shows that image backgrounds carry predictive signal and that ROI restriction sometimes helps; and it offers group-robustness, invariance, erasure, explanation-guided and generative mitigations, none of which is uniformly better than standard training. Every masking and background-removal study we found assumes that the shortcut lies outside the ROI. Reversed tests, which train on one artifact–label correlation and test on its opposite, are established, and synthetic biases have been moved between positions outside the disease region; no study places the same artifact inside versus outside the ROI used for masking and evaluates each placement on correlated, reversed and clean tests. The clearest report that removing one shortcut can amplify another comes from natural images. The review ends with a gap analysis that positions the thesis against the closest prior work and states what is and is not new.

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1 Purpose and review questions

Deep classifiers for medical images often rely on features that predict the label in the training data without being evidence of disease—*shortcuts* (Geirhos et al., 2020)[†]. A common defence is to segment the lesion or organ and classify only the region of interest. The thesis asks when that defence works, and finds that it depends on where the artifact sits: masking removes an artifact outside the ROI but keeps, and makes the model rely more on, an artifact inside it. This review sets out what was known before and alongside that work. It is organised by six questions:

RQ1 What evidence is there that medical image classifiers use content that is not pathology (artifacts, devices, acquisition signals, demographic correlates)? (Sec. 3)

RQ2 Which artifacts are documented in each modality, and where do they lie relative to the ROI? (Sec. 4)

RQ3 What is known about restricting a model to the ROI by masking, segmentation or background removal? (Sec. 5)

RQ4 How are shortcuts detected and audited? (Sec. 6)

RQ5 Which mitigation methods exist, and what does each require? (Sec. 7)

RQ6 What theory explains which feature a model uses, and how should claims of this kind be evaluated? (Secs. 8, 9)

Section 10 draws the gap analysis and Section 11 lists every included study.

2 Method

Search Searches were run in September and October 2026 with web and scholarly search engines, PubMed, the publishers' sites and the proceedings of MICCAI (Springer LNCS), CVF open ac-

cess (CVPR, ICCV), ECVA (ECCV), PMLR (ICML, AISTATS, MIDL), NeurIPS and OpenReview (ICLR). Query terms combined *shortcut learning, spurious correlation, Clever Hans, confounding, dataset bias* with *medical imaging, dermoscopy, skin lesion, ultrasound, calipers, chest radiograph, chest drain, chest tube, capsule endoscopy* and with *masking, segmentation, background, region of interest, counterfactual, invariance, concept erasure, group robustness.* We then followed references backwards and citations forwards from four anchor papers (DeGrave et al., 2021; Brown et al., 2023; Jiménez-Sánchez et al., 2023; Sourget et al., 2026).

Inclusion A study was included if it (i) was published, in final peer-reviewed form, between January 2021 and October 2026; (ii) appeared in a journal or main-track conference of recognised standing; and (iii) addressed one of RQ1–RQ6. For reading convenience we mark venues as **A** (Nature, Lancet and Science families, Medical Image Analysis, Radiology: Artificial Intelligence, JAMIA, Information Fusion; NeurIPS, ICML, ICLR, CVPR, ECCV, AAAI, AISTATS and MICCAI) or **B** (other established peer-reviewed journals and conferences, e.g. European Radiology, European Journal of Cancer, JMIR, Computers in Biology and Medicine, Machine Learning, Patterns, MIDL, ISBI, MLSys). The tiers are an aid for the reader, not a formal ranking.

Exclusion Preprints without a peer-reviewed version, workshop papers and papers in journals without a consistent standing in the field were excluded even when relevant. They are listed in Appendix A so that the reader can see what was left out and why, and are not counted; two of them (the source of a test design and the source of annotations the thesis uses) are also cited in the text. Studies of segmentation accuracy alone, and fairness studies without a bearing on the mechanism of shortcut use, were excluded. A small number of works published before 2021 are cited as background; they are marked with † and are not counted.

Verification For every included study the title, authors, venue, volume, pages or article number and DOI were checked against the publisher's or proceedings' own page. Each summary below states only what the paper's abstract or main results report; where we draw an inference of our own, the text says so.

Limitations of the method This is a structured narrative review by one reviewer, not a PRISMA systematic review: screening counts were not recorded, only English-language work was searched, and coverage of 2026 is necessarily incomplete because indexing lags publication. Statements of the form "we found no study" mean no study within this search.

3 Evidence that medical classifiers learn shortcuts (RQ1)

3.1 Confounded acquisition and dataset assembly

The early warning came from pneumonia detection, where models identified the hospital of origin from radiographs (Zech et al., 2018)†, and from *hidden stratification*, where overall metrics conceal failures on clinically important subsets (Oakden-Rayner et al., 2020)†. The COVID-19 literature then made the problem unmistakable. DeGrave et al. (2021) showed with saliency maps and generative counterfactuals that radiographic COVID-19 detectors relied on laterality markers, patient positioning and other confounds of the source datasets rather than on lung pathology, and lost much of their performance on external data. A systematic review of 62 COVID-19 imaging models found none suitable for clinical use, citing bias in small datasets and the variability of large, internationally sourced ones (Roberts et al., 2021). Maguolo and Nanni (2021) classified COVID-19 radiographs whose lung fields had been blacked out, showing that the datasets carried label information outside the anatomy the task is about.

3.2 Demographic attributes as encoded shortcuts

Deep models predict self-reported race from radiographs, even from degraded images (Gichoya et al., 2022), and chest-radiograph classifiers underdiagnose under-served groups, most strongly at intersections such as Hispanic female patients (Seyyed-Kalantari et al., 2021). Disease models encode protected characteristics in their features (Glocker et al., 2023a), and a chest-radiograph foundation model showed sex- and race-related biases (Glocker et al., 2023b); expert-level vision–language foundation models underdiagnose marginalised groups more than radiologists do (Yang et al., 2025). Yang et al. (2024) found that models which encode demographics more strongly show larger fairness gaps, and that debiasing which is optimal in distribution does not transfer to new sites. Two methods test reliance rather than encoding: *shortcut testing* varies how strongly a sensitive attribute is encoded and measures the effect on fairness (Brown et al., 2023), and Ong Ly et al. (2024) estimate external performance without external data from a model's dependence on shortcuts. Jones et al. (2024) separate three causal mechanisms of dataset bias (prevalence, presentation and annotation) that look alike in the data but call for different mitigations, and Banerjee et al. (2023) review causes, evaluation and mitigation of shortcuts for radiologists. For foundation-model features, pooling data across institutions can hide shortcut-like behaviour that per-site evaluation reveals (Germani et al., 2026).

3.3 Controlled, synthetic bias

Several studies inject a known bias to study its effect. Stanley et al. (2024) generate synthetic brain images with known disease and bias effects, place the bias either near the disease region (putamen) or far from it (postcentral gyrus), both outside the disease region itself (left insula), train and test at a nominal 70/30 bias prevalence, and show that this choice changes the performance gaps and how well three mitigations work (reweighing gave the most consistent near-zero gaps); a follow-up locates where in a convolutional network the bias is encoded and finds that strong encoding does not always mean the model uses it (Stanley et al., 2025). Sun et al. (2023) contaminate chest radiographs with a tag, hyperintensities or an obstruction and find that explanation methods differ strongly in whether they reveal the spurious signal. Boland et al. (2024) add synthetic shortcuts such as a square, at a fixed or a random position, to Waterbirds, CheXpert and ISIC 2017 images, train with a perfect association and test on balanced data, and use prediction depth to locate the layers where the shortcut is used. Corbetta et al. (2026) inject simulated acquisition artifacts into fundus and mammography images at prevalences from 0 to 100% and evaluate vision-language models on clean test sets and on sets in which the artifact–class associations are cyclically inverted; balanced accuracy falls towards chance under inversion, and concept supervision helps only partly.

For the thesis. The traps of the thesis belong to this controlled tradition. Stanley et al. (2024) are the closest precedent for position: they move a synthetic bias between two places, both outside the disease region, and find that it matters; Boland et al. (2024) compare fixed and random positions. Neither moves the bias across an ROI, studies ROI masking or uses a real artifact. The thesis varies an artifact's position relative to the ROI that masking keeps, and measures how masking's benefit changes with it, with pasted and real artifacts.

4 Artifact shortcuts by modality (RQ2)

4.1 Dermoscopy

Dermoscopy has the richest artifact literature. Ruler scale bars changed the melanoma predictions of a market-approved classifier (Winkler et al., 2021), as surgical skin markings had earlier (Winkler et al., 2019)[†]. Explanation-based audits found colour calibration patches, rulers and other artifacts used by ISIC models (Anders et al., 2022; Pahde et al., 2023), and the approach was later extended to semi-automated bias annotation across four medical tasks (Pahde et al., 2025). Hair occluding the lesion degrades classifiers (Wang et al., 2024), and adding synthetic dark-corner artifacts during training helps more than removing real ones (Pewton et al., 2024). The ISIC collections contain

duplicate images across releases, which can leak between training and test sets (Cassidy et al., 2022); the ISIC 2020 set was released with patient-level metadata (Rotemberg et al., 2021). Performance falls on darker skin and uncommon diseases (Daneshjou et al., 2022). Lesion segmentation, the source of dermoscopy ROIs, is reviewed by Mirikharaji et al. (2023).

Location. Calibration patches and dark corners lie mostly outside the lesion; rulers usually lie near the image edge and markings around the lesion, sometimes overlapping its border; hair crosses it. The studies we reviewed report the effect of an artifact's presence, not of where it lies.

4.2 Ultrasound

In fetal ultrasound, calipers and on-screen text act as shortcuts even for segmentation (Lin et al., 2024). The public thyroid datasets used in the thesis (Gong et al., 2021; Hou et al., 2024) contain sonographer calipers, mostly on or around the nodule (our observation; the dataset papers do not study them). The closest work we found is a thyroid validation study that considered calipers as a possible confound and judged that differences in their appearance between scanners did not explain its performance difference (Weng et al., 2023). Within our search, no peer-reviewed study measures calipers as a *classification* shortcut in thyroid or ovarian ultrasound, or relates the caliper's position to the nodule ROI.

4.3 Chest radiography

Pneumothorax classifiers detect the chest tube that treats the condition. Rueckel et al. (2021) reduced this confounding by training with in-image annotations, and Jiménez-Sánchez et al. (2023) trained on CheXpert and tested on CheXpert and NIH images with and without drains: AUROC on NIH fell from 81.1% with drains to 65.5% without. Non-expert drain and tube annotations for 3,543 NIH images and 1,011 PadChest images were later released as NEATX (Cheplygina et al., 2025); the thesis uses them. A meta-analysis of deep pneumothorax detection reports performance comparable to physicians but a high risk of bias in most studies (Sugibayashi et al., 2023). Saliency maps localise chest pathology worse than radiologists (Saporta et al., 2022), which limits explanation-based audits. Diffusion-based editing allows stress tests for acquisition, manifestation and population shift without new data (Pérez-García et al., 2024). Bassi et al. (2024) penalise relevance outside segmentation masks during training to make chest radiograph classifiers robust to background bias. Their evaluation is the closest to the trap design of the thesis: with an artificial mark placed outside the lungs, they test on matched (100%), cyclically reversed (0%) and artifact-free COVID-19 sets, and compare ISNet with alternatives that include a U-Net that segments the lungs first. The mark is always outside the lungs, so masking is expected to remove it.

Location. A drain projects over the lung field, so for lung-field masks it is an in-ROI artifact. Lung masking therefore cannot remove it: this is the boundary case discussed in the thesis.

4.4 Capsule endoscopy and other modalities

Pahde et al. (2025) include a gastrointestinal abnormality task among their four. We found no peer-reviewed study of bubbles or debris as a classification shortcut in capsule endoscopy, a modality in which the artifact and the lesion can share the same pixels.

5 Restricting the model to the region of interest (RQ3)

5.1 Backgrounds carry signal

In natural images, ImageNet classifiers recognise objects from backgrounds alone, and more accurate models rely less on the background (Xiao et al., 2021); foreground and background sensitivity varies

Table 1: Artifacts documented as shortcuts (2021–2026), with their typical position relative to the ROI used for masking. "Position" is our reading of the artifact type, not a quantity reported by the cited studies.

Modality	Artifact	Typical position	Studies
Dermoscopy	ruler, scale bar, skin marking	mostly outside; markings can overlap the border	Winkler et al. (2021); Anders et al. (2022); Pahde et al. (2023)
	hair	crosses the lesion	Wang et al. (2024)
	dark corner, colour patch	outside	Pewton et al. (2024); Anders et al. (2022)
Ultrasound	calipers, on-screen text	on or around the ROI	Lin et al. (2024) (segmentation only)
Chest radiograph	chest tube or drain	inside the lung field	Rueckel et al. (2021); Jiménez-Sánchez et al. (2023)
	laterality marker, text, positioning	outside	DeGrave et al. (2021)
	source-dataset signal	everywhere, incl. outside	Maguolo and Nanni (2021); Sourget et al. (2026)
Capsule endoscopy	bubbles, debris	can overlap the lesion	none found

strongly with training and architecture (Moayeri et al., 2022). The same holds in medicine. COVID-19 radiographs remain classifiable with the lungs removed (Maguolo and Nanni, 2021), and in a systematic masking study on chest radiographs and fundus images, models trained on full images performed better on images *without* the ROI than on images that contained only the ROI (Sourget et al., 2026).

5.2 ROI restriction as a remedy

Training on segmented dermoscopy images improved balanced accuracy for classifiers trained on HAM10000 but not for those trained on ISIC (Maron et al., 2021). ISNet uses segmentation masks only at training time, penalising relevance outside them, and improves robustness to background bias on chest radiographs (Bassi et al., 2024); in-image annotations serve a similar purpose for chest tubes (Rueckel et al., 2021). Each of these remedies assumes that the shortcut lies outside the region that is kept.

5.3 Removing one shortcut can amplify another

Li et al. (2023) showed in natural images that when several shortcuts are present, mitigating one makes the model rely more on the others, and proposed a last-layer ensemble against it. In our reading, ROI masking is a mitigation of exactly this kind: it removes background shortcuts and, if an artifact remains inside the ROI, the same dilemma predicts greater reliance on it. Li et al. did not study masking or medical images.

For the thesis. The prior work supports both halves of the problem—backgrounds carry shortcuts, and masking often helps—but every ROI-restriction study assumes the shortcut is in the background. Within our search, no study varies the artifact's location relative to the ROI used for masking (the closest, Stanley et al., 2024, varies a synthetic bias's distance from the disease region without masking), and none tests whether masking increases reliance on an in-ROI artifact in medical images.

6 Detecting and auditing shortcuts (RQ4)

Explanations Spectral relevance analysis clusters explanation heatmaps to reveal Clever-Hans strategies in ImageNet and ISIC 2019 (Anders et al., 2022); Reveal-to-Revise turns this into an

iterative cycle of revealing, modelling, removing and re-evaluating a bias (Pahde et al., 2023, 2025). Explanations are not reliably faithful, however: detection of injected confounders differs widely between methods (Sun et al., 2023), saliency localises chest pathology worse than radiologists (Saporta et al., 2022), and shortcut dependence can be traced to particular layers (Boland et al., 2024). Our inference: a heatmap cannot separate an in-ROI artifact from the pathology it overlaps, because both occupy the same pixels.

Counterfactuals and editing Diffusion-based counterfactuals add or remove a suspected shortcut and measure the change in prediction, with a masking scheme that confines the edit to a small region (Weng et al., 2024); RadEdit edits chest radiographs under multiple masks to stress-test classifiers (Pérez-García et al., 2024); and DeGrave et al. (2021) used generative counterfactuals to confirm the COVID-19 confounds.

Statistical tests Shortcut testing (Brown et al., 2023), estimation of external performance from internal data (Ong Ly et al., 2024), artifact labels on public datasets (Jiménez-Sánchez et al., 2023) and per-subgroup evaluation (Yang et al., 2024) probe reliance without heatmaps.

For the thesis. The thesis uses the counterfactual idea in its simplest form, pasting a real artifact into the same image inside or outside the ROI, which yields a same-image test of location rather than of presence.

7 Mitigation (RQ5)

7.1 Group robustness

With group labels (artifact × class), GroupDRO (Sagawa et al., 2020)†, variance-of-risk penalties (V-REx; Krueger et al., 2021) and last-layer retraining on a group-balanced set (DFR; Kirichenko et al., 2023) are standard. Izmailov et al. (2022) showed that ERM features already contain much of the core information, so retraining only the last layer can recover robustness, which motivates linear heads on frozen features. Without group labels, JTT up-weights the errors of a first model (Liu et al., 2021) and AFR reweights the last layer automatically (Qiu et al., 2023). Large benchmarks temper expectations: across 20 algorithms and 12 datasets, methods improved subgroup robustness only for some types of shift (Yang et al., 2023), and in medical imaging, bias-mitigation algorithms did not significantly improve fairness over ERM, while model selection mattered more (Zong et al., 2023).

7.2 Auxiliary labels and invariance

Counterfactual invariance formalises what a shortcut-free predictor should satisfy; in the anti-causal setting, in which the label causes the image, it implies that the prediction is independent of the nuisance *given the label* (Veitch et al., 2021). Makar et al. (2022) enforce the conditional independences implied by the causal graph with auxiliary labels available at training time, and NuRD trains under a distribution in which the nuisance–label relation is broken (Puli et al., 2022). Kaur et al. (2023) show that the correct independence constraint depends on the data-generating process and that no single constraint suits all shifts, consistent with the mechanism-specific view of Jones et al. (2024).

7.3 Concept erasure

Linear erasure removes a concept from the features: iterative nullspace projection (Ravfogel et al., 2020)†, adversarial erasure (Ravfogel et al., 2022) and the closed-form LEACE (Belrose et al., 2023). When the concept is correlated with the label, erasure also removes label signal; SPLINCE preserves covariance with the task label while removing the concept (Holstege et al., 2025).

Table 2: Mitigation families and what they need. "Thesis arm" names the method of the thesis that belongs to each family (ERM: standard training; mask: ROI masking).

Family	Representative studies	Needs at training	Evidence in medical imaging	Thesis arm
ROI restriction	Maron et al. (2021); Bassi et al. (2024)	ROI masks	mixed; assumes background shortcut	mask
Group reweighting / DRO	Krueger et al. (2021); Kirichenko et al. (2023); Qiu et al. (2023)	group labels (AFR, JTT: none)	no consistent gain over ERM (Zong et al., 2023)	group balancing
Conditional invariance	Veitch et al. (2021); Makar et al. (2022); Puli et al. (2022)	auxiliary (artifact) labels	limited	class-conditional constraint
Concept erasure	Belrose et al. (2023); Holstege et al. (2025)	concept labels or pairs	limited	U-MtE (label-free)
Explanation-guided	Anders et al. (2022); Dreyer et al. (2024); Pahde et al. (2023)	concept annotation	dermoscopy, chest radiograph	—
Generative / data	Ktena et al. (2024); Weng et al. (2024)	a generative model	histopathology, chest radiograph, dermoscopy	—

7.4 Explanation-guided and data-centric methods

Class-artifact compensation (Anders et al., 2022), gradient penalties along a concept direction in latent space (Dreyer et al., 2024) and relevance penalties outside a mask (Bassi et al., 2024) use explanations to suppress a known bias. On the data side, in-image annotations (Rueckel et al., 2021), artifact augmentation (Pewton et al., 2024), diffusion-generated synthetic data, which improved fairness out of distribution in histopathology, chest radiography and dermatology (Ktena et al., 2024), and counterfactual removal of shortcuts (Weng et al., 2024) change what the model sees.

For the thesis. The thesis's recommended remedy, a class-conditional mean constraint on the masked features, is the first-moment, linear case of the conditional independence of Veitch et al. (2021) and Makar et al. (2022). What the thesis adds is its derivation from the location law, its use after masking, and a registered test of whether any remedy *dominates* masking; the benchmark results of Yang et al. (2023) and Zong et al. (2023) make it unsurprising that none does.

8 Theory of feature choice (RQ6, part 1)

Why does a model pick a shortcut when a stable feature is available? Nagarajan et al. (2021) show that even when the invariant feature fully predicts the label, geometric and statistical skews push ERM toward spurious ones. Gradient descent with cross-entropy can starve less salient features (Pezeshki et al., 2021), and the max-margin bias of default ERM produces shortcut reliance even when the shortcut adds no information (Puli et al., 2023). Hermann et al. (2024) separate a feature's *predictivity* from its *availability* (how easily it is extracted) and quantify shortcut bias as reliance on the more available but less predictive feature. Invariant risk minimisation can fail outside the linear regime or with few environments (Rosenfeld et al., 2021). For frozen features, linear probing can generalise better out of distribution than fine-tuning, which distorts good pretrained features (Kumar et al., 2022).

For the thesis. The thesis's linear-Gaussian account fits this line of work. Masking removes competing evidence outside the ROI, which in Hermann et al.'s terms changes the relative predictivity of what remains, so reliance on an in-ROI artifact grows. Unlike the explanatory theories above, the account was used to register quantitative predictions before the results existed.

9 Evaluation methodology (RQ6, part 2)

Methodological failures in medical-imaging machine learning, from dataset bias to publication incentives, are reviewed by Varoquaux and Cheplygina (2022), and data leakage, including non-independent splits, inflated results in at least 294 papers across 17 fields (Kapoor and Narayanan, 2023). Variance from data sampling, initialisation and hyperparameters changes benchmark conclusions, and accounting for several sources at once estimates it better (Bouthillier et al., 2021). In MICCAI papers, confidence intervals are rarely reported and their typical width exceeds the gap between the first- and second-ranked methods (Christodoulou et al., 2024). Metrics should follow from the clinical question (Maier-Hein et al., 2024), and reporting should follow CLAIM (Tejani et al., 2024). Model selection can matter more than the mitigation algorithm (Zong et al., 2023; Yang et al., 2023), and subgroup failures are hidden by overall metrics (Yang et al., 2024), as hidden stratification had shown (Oakden-Rayner et al., 2020)[†].

Reversed tests. Training on one cue–label correlation and testing on its reversal is an established design. Colored MNIST trains at 80–90% colour–label agreement and tests at 10% (Arjovsky et al., 2019). In medical images, Puli et al. (2022) train chest-radiograph models at 90% agreement between hospital source and label and test at 10% (and at 30% and 50%), the same 90/10 schedule as the thesis; Bassi et al. (2024) use matched, reversed and artifact-free tests; Corbetta et al. (2026) invert artifact–class associations cyclically; and a workshop study built trap sets that invert seven dermoscopy artifact associations and evaluated lesion-exterior masking (Appendix A). The thesis takes this design from them; it does not claim it.

For the thesis. These studies motivate the evaluation choices of the thesis: patient- or lesion-grouped splits; a bootstrap that resamples training seeds and test images jointly, since a test set shared by several seeds is one sample, not several; Holm-corrected families; registration before results; correlated, reversed and clean tests as in the reversed-test literature above; and evaluation on shortcut-conflicting cases as well as on the whole unaltered test set.

10 Gap analysis: where the thesis stands

Table 3 sets each contribution of the thesis against the closest peer-reviewed work found.

Table 3: Gap analysis. "Open" means not addressed by any included study.

Question	Established in 2021–2026	Open	Thesis contribution
Where does the shortcut sit?	Artifacts are documented by presence (Winkler et al., 2021; Rueckel et al., 2021; Lin et al., 2024; Wang et al., 2024); backgrounds carry signal (Maguolo and Nanni, 2021; Sourget et al., 2026; Xiao et al., 2021); a synthetic bias's distance from the disease region changes performance gaps (Stanley et al., 2024)	Location relative to the masking ROI as an experimental variable, with real artifacts	Overlap r controlled in sweeps; real traps with in-ROI and out-of-ROI artifacts; transplant of the same real artifact across the ROI boundary of the same images

Question	Established in 2021–2026	Open	Thesis contribution
Does ROI masking fix it?	Sometimes (Maron et al., 2021; Bassi et al., 2024); every study assumes a background shortcut	When and why masking fails	Location law: masking removes out-of-ROI shortcuts and leaves in-ROI ones, in four cohorts and three modalities
Is the test design new?	No. Reversed tests: Arjovsky et al. (2019); Puli et al. (2022); Bassi et al. (2024); Corbetta et al. (2026); synthetic bias moved between positions: Stanley et al. (2024); Boland et al. (2024)	Crossing the artifact's position relative to the ROI with masking, under correlated, reversed and clean tests	The same artifact inside vs outside the ROI, with and without masking, each on three tests; real as well as pasted artifacts
Can a mitigation amplify a shortcut?	Yes, for multiple shortcuts in natural images (Li et al., 2023)	Whether masking amplifies an in-ROI artifact in medical images, and by what mechanism	Reliance on the in-ROI artifact grows after masking because competing evidence is removed
Can it be predicted?	Explanatory theories of feature choice (Nagarajan et al., 2021; Puli et al., 2023; Hermann et al., 2024)	Quantitative, prospectively tested predictions	Two-parameter linear-Gaussian account; predictions registered before the results
Is harm visible clinically?	Overall metrics hide subgroup failures (Yang et al., 2024); see also Oakden-Rayner et al. (2020)†	Effect of masking on shortcut-conflicting cases of unaltered data	Lower AUROC on conflicting cases of patient-disjoint and new-patient thyroid data; sensitivity loss among malignant nodules with calipers
What to use instead?	No method uniformly beats ERM (Yang et al., 2023; Zong et al., 2023); conditional invariance with auxiliary labels (Veitch et al., 2021; Makar et al., 2022); erasure (Belrose et al., 2023; Holstege et al., 2025)	Remedies tested against masking under a registered criterion	42-component dominance criterion, two registered rounds; no remedy dominates; the class-conditional constraint comes closest
Which modalities?	Mostly dermoscopy and chest radiography; calipers in fetal-US segmentation (Lin et al., 2024)	Thyroid and ovarian ultrasound calipers, capsule debris as classification shortcuts	Traps in thyroid and ovarian ultrasound and capsule endoscopy, alongside dermoscopy
How strong is the evidence?	CIs rarely reported (Christodoulou et al., 2024); seed and data variance (Bouthillier et al., 2021); leakage (Kapoor and Narayanan, 2023)	Joint seed × image uncertainty for shortcut experiments	Crossed bootstrap, Holm families, registration, every interval traced to a result file

What is not new That a mask cannot remove what it keeps is self-evident; what was not established is that masking makes reliance on a kept artifact *grow*, how large this is on real artifacts, that it is caused by location, and how to predict it. The correlated, reversed and clean test design is borrowed (Arjovsky et al., 2019; Puli et al., 2022; Bassi et al., 2024; Corbetta et al., 2026), and varying a synthetic bias's position is not new (Stanley et al., 2024; Boland et al., 2024); what is new is crossing the artifact's position relative to the ROI with ROI masking under that design, and doing so with real artifacts. Group balancing, last-layer retraining, IRM and V-REx are established methods used as baselines. The class-conditional constraint is a special case of known conditional-invariance regularisers (Veitch et al., 2021; Makar et al., 2022), and disease-preserving erasure overlaps with SPLINCE (Holstege et al., 2025); the thesis presents both as adaptations, not as new methods.

Threats to this positioning (i) Two workshop papers study masking directly: early versus late background masking for fine-grained recognition, and artifact-based domain generalisation with test-time masking in dermoscopy (Appendix A). Both assume background shortcuts, but a reviewer may know them. A third workshop paper, Bissoto et al. (2020), already combines trap sets that reverse artifact associations with lesion-exterior masking in dermoscopy; it does not vary where the artifact lies relative to the lesion, but it is the closest single precedent for the test design and must be cited. (ii) The whac-a-mole dilemma (Li et al., 2023) anticipates the amplification conceptually; the thesis should cite it and state that it supplies the location-specific mechanism and the medical evidence. (iii) Coverage of 2026 is incomplete; the review should be repeated before submission.

11 Summary of included studies

Table 4: All 70 included studies (2021–2026), grouped by theme. Tier: A or B as defined in Sec. 2. CXR: chest radiograph; US: ultrasound; NI: natural images; –: not modality-specific.

Study	Venue	Tier	Modality	Finding or contribution	Relevance to the thesis
Evidence of shortcut learning and bias					
DeGrave et al. (2021)	Nat. Mach. Intell.	A	CXR	COVID-19 detectors use dataset confounds	motivating example
Roberts et al. (2021)	Nat. Mach. Intell.	A	CXR, CT	none of 62 models clinically usable	dataset bias
Maguolo and Nanni (2021)	Inf. Fusion	A	CXR	classification with lungs blacked out	signal outside the ROI
Seyyed-Kalantari et al. (2021)	Nat. Med.	A	CXR	underdiagnosis of under-served groups	hidden subgroup harm
Gichoya et al. (2022)	Lancet Digit. Health	A	multiple	race predictable from images	invisible correlates
Glocker et al. (2023a)	eBioMedicine	A	CXR	disease models encode protected attributes	encoding vs use
Glocker et al. (2023b)	Radiol. AI	A	CXR	bias in a foundation model	frozen-feature heads
Brown et al. (2023)	Nat. Commun.	A	derm., CXR	shortcut testing	test of reliance
Banerjee et al. (2023)	J. Am. Coll. Radiol.	B	radiology	review of causes and mitigation	background
Yang et al. (2024)	Nat. Med.	A	multiple	fairness does not transfer to new sites	evaluation on new data
Yang et al. (2025)	Sci. Adv.	A	CXR	VLM foundation models underdiagnose	foundation models
Jones et al. (2024)	Nat. Mach. Intell.	A	–	three causal bias mechanisms	choice of remedy
Ong Ly et al. (2024)	npj Digit. Med.	A	medical	generalisation estimated without external data	reliance estimation

Study	Venue	Tier	Modality	Finding or contribution	Relevance to the thesis
Stanley et al. (2024)	JAMIA	A	brain MRI	synthetic bias near or far from the disease region; distance matters; reweighing best	closest location precedent
Stanley et al. (2025)	eBioMedicine	A	brain MRI	where bias is encoded	encoding vs use
Sun et al. (2023)	MICCAI	A	CXR	explanations vary in detecting injected confounders	limits of heatmaps
Boland et al. (2024)	MIDL	B	CXR, derm., NI	fixed vs random shortcut position; layer-wise localisation	position precedent
Corbetta et al. (2026)	MICCAI	A	fundus, mammo.	inverted and clean tests for injected artifacts	reversed-test precedent
Sourget et al. (2026)	J. Imaging Inform. Med.	B	CXR, fundus	ROI-removed images remain classifiable	closest masking study
Germani et al. (2026)	Med. Image Anal.	A	medical	pooled data hide shortcut-like behaviour	foundation features
Ktena et al. (2024)	Nat. Med.	A	histo., CXR, derm.	synthetic data improve OOD fairness	generative remedy
Dermoscopy					
Winkler et al. (2021)	Eur. J. Cancer	B	derm.	scale bars alter a market-approved CNN	artifact by presence
Maron et al. (2021)	JMIR	B	derm.	segmentation helps on HAM, not ISIC	masking evidence (mixed)
Daneshjou et al. (2022)	Sci. Adv.	A	derm. (clinical)	lower accuracy on dark skin	distribution shift
Cassidy et al. (2022)	Med. Image Anal.	A	derm.	ISIC duplicates and benchmarks	grouped splits
Mirikharaji et al. (2023)	Med. Image Anal.	A	derm.	survey of lesion segmentation	source of ROI masks
Wang et al. (2024)	Comput. Biol. Med.	B	derm.	hair degrades classifiers	in-ROI artifact
Pewton et al. (2024)	Comput. Methods Programs Biomed.	B	derm.	adding dark corners beats removal	out-of-ROI artifact
Anders et al. (2022)	Inf. Fusion	A	derm., NI	SpRAy and class-artifact compensation	detection and remedy
Pahde et al. (2023)	MICCAI	A	derm., X-ray	Reveal-to-Revise life cycle	detection and remedy
Pahde et al. (2025)	Mach. Learn.	B	4 tasks	semi-automated bias annotation	artifact labels
Ultrasound and chest radiography					
Lin et al. (2024)	MICCAI	A	US	calipers and text as segmentation shortcuts	ultrasound calipers
Weng et al. (2023)	Clin. Imaging	B	US	thyroid model validated on other scanners; caliper differences judged not to explain results	calipers as a confound
Rueckel et al. (2021)	Eur. Radiol.	B	CXR	annotations reduce chest-tube confounding	in-ROI device
Sugibayashi et al. (2023)	Eur. Respir. Rev.	B	CXR	meta-analysis; high risk of bias	drain context
Jiménez-Sánchez et al. (2023)	IEEE ISBI	B	CXR	drain labels expose a shortcut	chest boundary case
Saporta et al. (2022)	Nat. Mach. Intell.	A	CXR	saliency localises worse than radiologists	limits of heatmaps
Pérez-García et al. (2024)	ECCV	A	CXR	diffusion editing for stress tests	counterfactual testing
Bassi et al. (2024)	Nat. Commun.	A	CXR	relevance penalties outside masks; matched, reversed and clean tests	mask-based remedy; test-design precedent

Study	Venue	Tier	Modality	Finding or contribution	Relevance to the thesis
Backgrounds and masking in natural images					
Xiao et al. (2021)	ICLR	A	NI	backgrounds alone predict the class	background shortcuts
Moayeri et al. (2022)	CVPR	A	NI	foreground/background sensitivity	masking evaluation
Li et al. (2023)	CVPR	A	NI	mitigating one shortcut amplifies others	closest to amplification
Detection by counterfactuals and explanation-guided training					
Weng et al. (2024)	ECCV	A	NI, medical	diffusion counterfactuals add/remove shortcuts	same-image test
Dreyer et al. (2024)	AAAI	A	derm., NI	latent gradient penalisation	explanation remedy
Mitigation: group robustness and benchmarks					
Krueger et al. (2021)	ICML	A	–	V-REx	baseline
Liu et al. (2021)	ICML	A	–	JTT, no group labels	label-free baseline
Izmailov et al. (2022)	NeurIPS	A	–	ERM features hold core information	frozen features
Kirichenko et al. (2023)	ICLR	A	–	last-layer retraining (DFR)	balanced head
Qiu et al. (2023)	ICML	A	–	automatic feature reweighting	label-free baseline
Yang et al. (2023)	ICML	A	mixed	no method best across shifts	no dominance expected
Zong et al. (2023)	ICLR	A	multiple	mitigation ≈ ERM in medicine	no dominance expected
Mitigation: invariance and erasure					
Veitch et al. (2021)	NeurIPS	A	–	counterfactual invariance	basis of the constraint
Makar et al. (2022)	AISTATS	A	–	auxiliary-label conditional independence	closest remedy
Puli et al. (2022)	ICLR	A	NI, CXR	nuisance-randomised distillation	alternative remedy
Kaur et al. (2023)	ICLR	A	–	constraint depends on the causal graph	why class-conditional
Ravfogel et al. (2022)	ICML	A	–	adversarial linear erasure	erasure family
Belrose et al. (2023)	NeurIPS	A	–	closed-form linear erasure	erasure baseline
Holstege et al. (2025)	NeurIPS	A	–	erasure preserving task signal	disease-preserving erasure
Theory					
Nagarajan et al. (2021)	ICLR	A	–	skews drive spurious reliance	theory
Rosenfeld et al. (2021)	ICLR	A	–	limits of IRM	IRM baseline
Pezeshki et al. (2021)	NeurIPS	A	–	gradient starvation	theory
Kumar et al. (2022)	ICLR	A	–	linear probing vs fine-tuning OOD	frozen heads
Puli et al. (2023)	NeurIPS	A	–	max-margin causes shortcuts	theory
Hermann et al. (2024)	ICLR	A	–	availability vs predictivity	closest to the account
Evaluation methodology and reporting					
Varoquaux and Cheplygina (2022)	npj Digit. Med.	A	–	methodological failures	study design
Kapoor and Narayanan (2023)	Patterns	B	–	leakage taxonomy	grouped splits
Bouthillier et al. (2021)	MLSys	B	–	sources of benchmark variance	crossed bootstrap
Christodoulou et al. (2024)	MICCAI	A	–	CIs rarely reported	intervals everywhere

Study	Venue	Tier	Modality	Finding or contribution	Relevance to the thesis
Maier-Hein et al. (2024)	Nat. Methods	A	–	problem-driven metrics	metric choice
Tejani et al. (2024)	Radiol. AI	A	–	CLAIM 2024 checklist	reporting

A Identified but excluded by the venue criterion

These items are relevant but do not meet the venue criterion of Sec. 2: preprints, workshop papers or journals outside it. They are listed for transparency and are not counted.

- Aniraj, Dantas, Ienco, Marcos (2023), *Masking strategies for background bias removal in computer vision models*, ICCV Workshops (OODCV): early (image-level) masking gave the best out-of-distribution accuracy in fine-grained recognition. The closest general-vision masking study; workshop paper.
- Bissoto, Valle, Avila (2020), *Debiasing skin lesion datasets and models? Not so fast*, CVPR Workshops, pp. 3192–3201: trap sets that reverse seven dermoscopy artifact associations, evaluated with lesion-exterior normalisation. Before 2021 and a workshop paper, but the closest precedent for the reversed test with masking.
- Bissoto, Barata, Valle, Avila (2022), *Artifact-based domain generalization of skin lesion models*, ECCV Workshops (ISIC); and Bissoto et al. (2023), *Test-time selection for robust skin lesion analysis*, MICCAI Workshops (ISIC). Workshop papers.
- Boland et al. (2024), *All you need is a guiding hand: mitigating shortcut bias in deep learning models for medical imaging*, MICCAI FAIMI workshop (LNCS 15198).
- Nauta, Walsh, Dubowski, Seifert (2022), *Uncovering and correcting shortcut learning in machine learning models for skin cancer diagnosis*, *Diagnostics* (MDPI).
- Jin, Gerych, Ghassemi (2024), *MaskMedPaint*, arXiv:2411.10686 (preprint).
- Pedersen, Sydendal, Cheplygina, Sourget (2026), real-world chest X-ray shortcuts in MedCLIP, arXiv:2608.12086 (preprint).
- Kina, Petersen (2026), *Prevalence calibration as shortcut mitigation*, arXiv:2609.07922 (preprint).
- Chuang et al. (2023), *Debiasing vision-language models via biased prompts*, arXiv:2302.00070 (preprint).
- Arjovsky, Bottou, Gulrajani, Lopez-Paz (2019), *Invariant risk minimization*, arXiv:1907.02893: a preprint and before 2021, but the source of the IRMv1 baseline and of the Colored MNIST reversed test (Arjovsky et al., 2019).
- Cheplygina et al. (2025), *Augmenting chest X-ray datasets with non-expert annotations*, MIUA, LNCS 15916: the NEATX drain annotations used by the thesis (Cheplygina et al., 2025); cited for data provenance.

B Datasets and encoders cited for provenance

Cited for the data and models the thesis uses, not as evidence: ISIC 2020 (Rotemberg et al., 2021), TN3K (Gong et al., 2021), the thyroid pathology dataset (Hou et al., 2024), DINOv2 (Oquab et al., 2024), RAD-DINO (Pérez-García et al., 2025) and PanDerm (Yan et al., 2025).

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